LAPORAN PRAKTIKUM

Mata Kuliah Basis Data

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Kelas: SE-08-02

Asprak:

PROGRAM STUDI S1 SOFTWARE ENGINEERING

TELKOM UNIVERSITY PURWOKERTO

APRIL 2026

MODUL 8
JURNAL

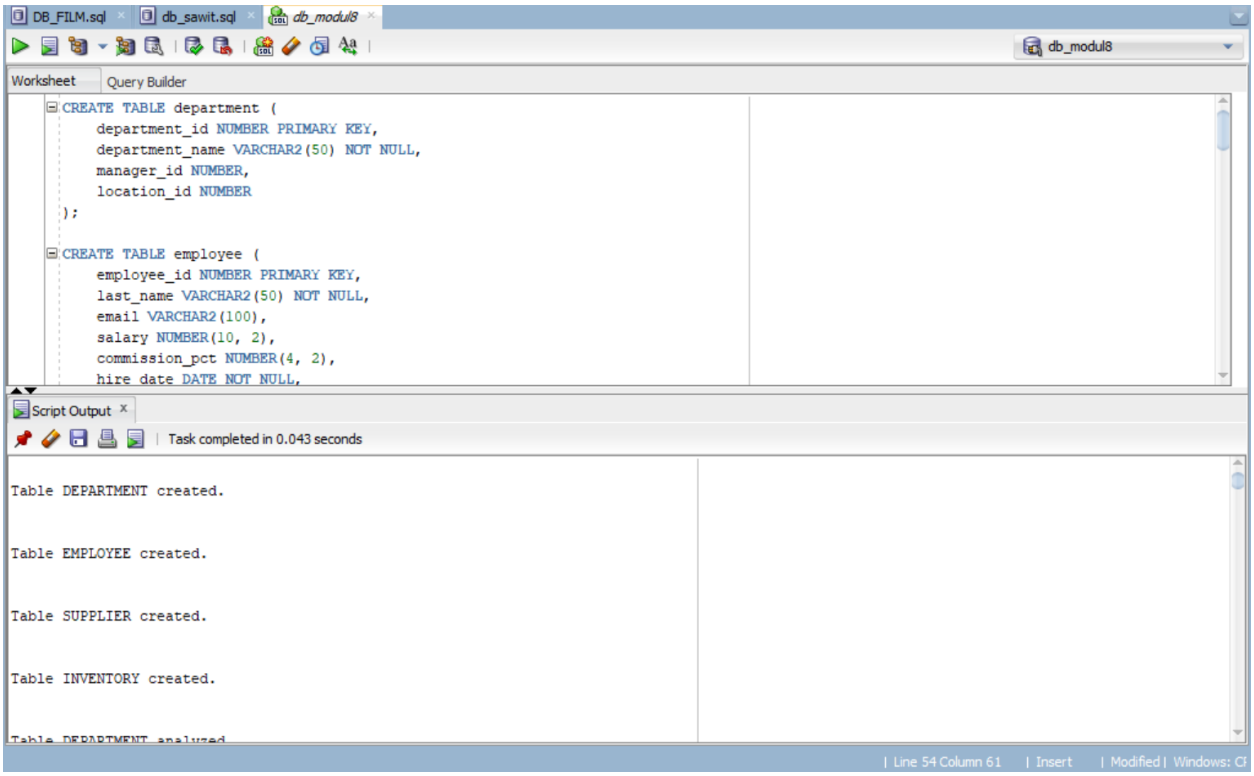
Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

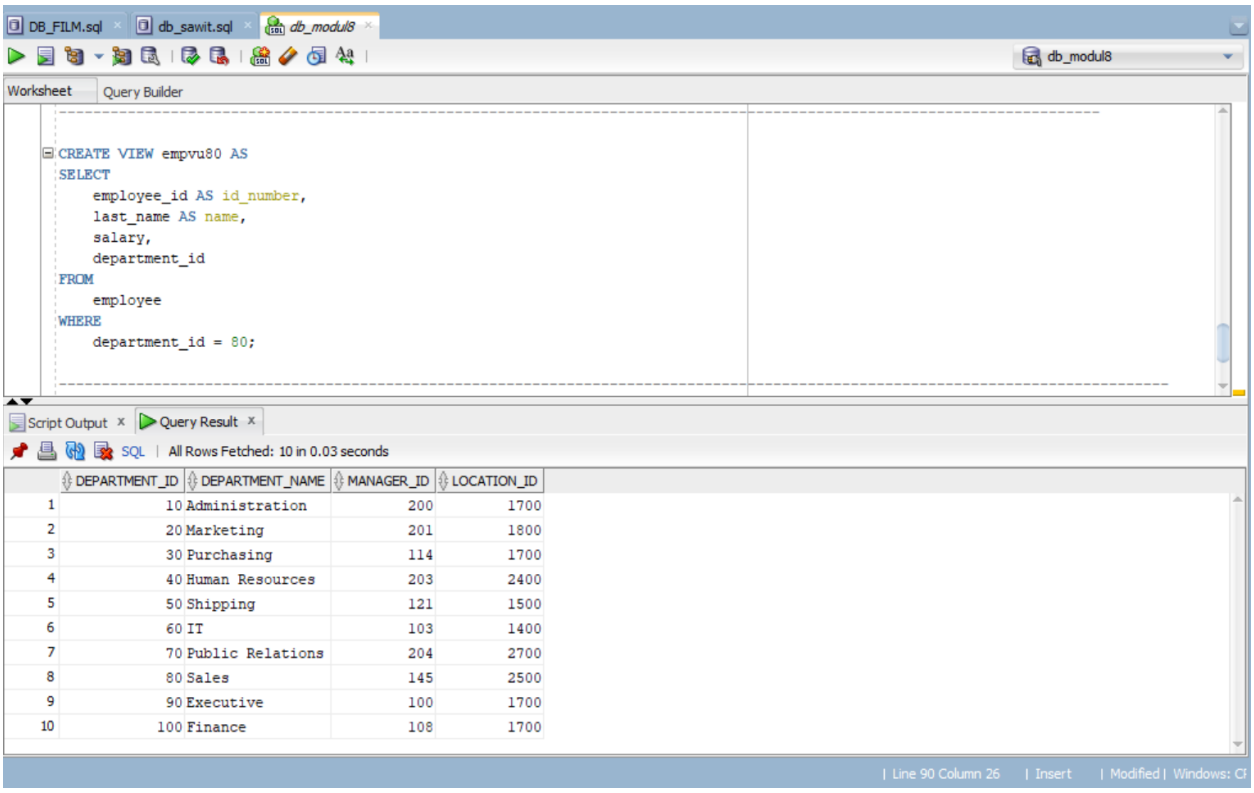
8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawaban :

1. Create Table





DB_FILM.sql | db_sawit.sql | db_modul8

db_modul8

Worksheet | Query Builder

```
INSERT INTO department VALUES (80, 'Sales', 145, 2500); -- soal No. 5
INSERT INTO department VALUES (90, 'Executive', 100, 1700);
INSERT INTO department VALUES (100, 'Finance', 108, 1700);

INSERT INTO employee VALUES (101, 'Smith', 'smith@mail.com', 4000, NULL, TO_DATE('2023-01-15', 'YYYY-MM-DD'), 20);
INSERT INTO employee VALUES (102, 'Johnson', 'johnson@mail.com', 5500, NULL, TO_DATE('2022-03-22', 'YYYY-MM-DD'), 60);
INSERT INTO employee VALUES (103, 'Williams', 'williams@mail.com', 8000, 0.15, TO_DATE('2021-07-01', 'YYYY-MM-DD'), 80); -- Dept 80
INSERT INTO employee VALUES (104, 'Jones', 'jones@mail.com', 4500, NULL, TO_DATE('2023-05-10', 'YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (105, 'Brown', 'brown@mail.com', 7500, 0.20, TO_DATE('2020-11-18', 'YYYY-MM-DD'), 80); -- Dept 80
INSERT INTO employee VALUES (106, 'Davis', 'davis@mail.com', 6000, NULL, TO_DATE('2022-09-05', 'YYYY-MM-DD'), 90);
INSERT INTO employee VALUES (107, 'Miller', 'miller@mail.com', 7200, 0.10, TO_DATE('2021-02-28', 'YYYY-MM-DD'), 80); -- Dept 80
INSERT INTO employee VALUES (108, 'Wilson', 'wilson@mail.com', 4200, NULL, TO_DATE('2023-08-14', 'YYYY-MM-DD'), 30);
INSERT INTO employee VALUES (109, 'Moore', 'moore@mail.com', 5100, NULL, TO_DATE('2022-12-01', 'YYYY-MM-DD'), 100);
INSERT INTO employee VALUES (110, 'Taylor', 'taylor@mail.com', 7800, 0.25, TO_DATE('2019-04-10', 'YYYY-MM-DD'), 80); -- Dept 80
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.016 seconds

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	101	Smith	smith@mail.com	4000	(null)	15-JAN-23	20
2	102	Johnson	johnson@mail.com	5500	(null)	22-MAR-22	60
3	103	Williams	williams@mail.com	8000	0.15	01-JUL-21	80
4	104	Jones	jones@mail.com	4500	(null)	10-MAY-23	50
5	105	Brown	brown@mail.com	7500	0.2	18-NOV-20	80
6	106	Davis	davis@mail.com	6000	(null)	05-SEP-22	90
7	107	Miller	miller@mail.com	7200	0.1	28-FEB-21	80
8	108	Wilson	wilson@mail.com	4200	(null)	14-AUG-23	30
9	109	Moore	moore@mail.com	5100	(null)	01-DEC-22	100
10	110	Taylor	taylor@mail.com	7800	0.25	10-APR-19	80

| Line 91 Column 24 | Insert | Modified | Windows: CF

```
-------------------------------------------------------------

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;

-------------------------------------------------------------
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.016 seconds

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	101	Smith	smith@mail.com	4000	(null)	15-JAN-23	20
2	102	Johnson	johnson@mail.com	5500	(null)	22-MAR-22	60
3	103	Williams	williams@mail.com	8000	0.15	01-JUL-21	80
4	104	Jones	jones@mail.com	4500	(null)	10-MAY-23	50
5	105	Brown	brown@mail.com	7500	0.2	18-NOV-20	80
6	106	Davis	davis@mail.com	6000	(null)	05-SEP-22	90
7	107	Miller	miller@mail.com	7200	0.1	28-FEB-21	80
8	108	Wilson	wilson@mail.com	4200	(null)	14-AUG-23	30
9	109	Moore	moore@mail.com	5100	(null)	01-DEC-22	100
10	110	Taylor	taylor@mail.com	7800	0.25	10-APR-19	80

| Line 91 Column 24 | Insert | Modified | Windows: CF

Oracle SQL Developer : db_modul8

File Edit View Navigate Run Source Team Tools Window Help

DBA

Connections
- db_film
- db_modul8
  - Container Database
  - Database Configuration
  - Database Status
  - Data Pump
  - Performance
  - RMAN Backup/Recovery
  - Resource Manager
  - SQL Translator Framework
  - Scheduler
  - Security
  - Storage
  - Tuning

DB_FILM.sql | db_sawit.sql | db_modul8

Worksheet | Query Builder

```sql
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM
    employee
WHERE
    department_id = 80;

---------------------------------------------------------------------------------------------------

DESCRIBE empvu80;
```

Script Output | Query Result

Task completed in 0.04 seconds

```
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(50)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER
Name          Null?    Type
------------- -------- ------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(50)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER
```

Line 86 Column 18 | Insert | Modified | Windows: CI

Oracle SQL Developer : db_modul8

File Edit View Navigate Run Source Team Tools Window Help

DBA

Connections
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DB_FILM.sql | db_sawit.sql | db_modul8

Worksheet | Query Builder

```sql
    last_name AS name,
    salary,
    department_id
FROM
    employee
WHERE
    department_id = 80;

---------------------------------------------------------------------------------------------------

DESCRIBE empvu80;

CREATE INDEX emp_department_idx ON employee (department_id);

SELECT * FROM department;
SELECT * FROM employee;
```

Script Output | Query Result

Task completed in 0.043 seconds

```
View EMPVU80 created.

Name          Null?    Type
------------- -------- ------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(50)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER

Index EMP_DEPARTMENT_IDX created.
```

Line 85 Column 1 | Insert | Modified | Windows: CI